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APPLIED RESEARCH

AI-Enhanced Quantum-Secured IoT Communication Framework for 6G Cognitive Radio Networks

P. DEEPANRAMKUMAR AND A. HELEN SHARMILA
School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh 522237, India
Corresponding author: P. Deepanramkumar (deepanramkumar.p@vitap.ac.in)

ABSTRACT The advent of 6G wireless communication promises improvements in signal coverage, data rates, and latency, addressing increased connectivity demands due to the proliferation of 5G, IoT, and augmented reality. This paper introduces a Quantum-Secured IoT Communication Framework designed for 6G Cognitive Radio Networks (CRNs) in order to cater to the growing need for dependable and protected connectivity. Noteworthy aspects of this framework encompass dual-layer authentication utilizing Quantum Key Distribution (QKD) and Public Key Infrastructure (PKI), secured spectrum access regulations, and efficient beamforming strategies. Moreover, the framework employs a Reinforcement Learning-based Ensemble Regression (RL-ER) model for spectrum sensing and a Multi-Layer Perceptron with Kalman Filter (MLP-KF) for Channel State Information (CSI) prediction. Simulations demonstrate that the framework significantly improves prediction accuracy, encryption and decryption times, and error rates, thereby enhancing IoT network performance with better signal coverage, reduced latency, and robust security.

INDEX TERMS 6G wireless communication, cognitive radio networks (CRNs), quantum key distribution (QKD), reinforcement learning-based ensemble regression (RL-ER), multi-layer perceptron with Kalman filter (MLP-KF).

I. INTRODUCTION

Since the introduction of 6G into wireless communication, there have been rapid improvements, more precisely in the key performance indicators regarding latency, data transmission, and signal coverage. This need for reliable connectivity and efficient data transfer to wireless devices has driven the growth of these technologies forward and continues through the advance of technologies using 5G, the Internet of Things, and enhanced reality [1]. This is how the IoT changes the interaction with sensors by smart devices in quite a number of fields, ranging from homes to industrial automation. Its growth has shown that the technology associated with spectral management, its data traffic, and connection needs need better solutions [2]. This makes necessary the establishment of numerous sensors and devices. Dynamic spectrum access, with some of the latest technologies such as cognitive radio, is under consideration to help improve the efficiency of spectra. With dynamic spectrum access, unlicensed users get access to the idle spectrum, thus improving its utilization. Cognitive radio, generally proven as an effective solution for solving the issue of scarcity of spectrum, leads to the emergence of cognitive radio sensor networks as a possible solution to reduce congestion in the unlicensed frequency bands and high spectrum demands [3].

Recent technological advancements have underscored the significance of real-time spectrum sensing in dynamic spectrum environments, attracting considerable research interest. Beamforming is crucial in this context as it ensures reliable connectivity by directing signals towards specific devices, especially in scenarios with limited wireless resources constrained by routing requirements. Secure routing protocols based on optimization algorithms have been created to guarantee reliable message delivery while maintaining high standards of security and privacy. Integrating cognitive radio

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with spread spectrum modulation techniques has proven to be an effective method for creating secure communication channels that are resistant to narrowband jamming and other interference tactics [4]. This integration facilitates the dynamic adjustment of transceiver parameters based on available channels, allowing multiple wireless communication systems to operate concurrently. Moreover, it is anticipated that data rates and network capacity will be enhanced to a great extent by resorting to massive MIMO technology, deploying large antenna arrays at both the transmission and reception ends. The integration of massive MIMO into 6G networks is expected to greatly enhance the performance of IoT networks in providing signaling coverage and reducing latencies [5]. Further, blockchain technology is recognized for its decentralized and distributed ledger system, as it is emerging as a solution to address the securities, privacy, and trust of wireless communications and IoT applications [15].

A. MOTIVATIONS & OBJECTIVES

The design of 6G CIoT networks will have to accommodate the use of the spectrum so as to allow safe and effective communication. Some of the main motivating factors are:

1) Security Issues: With the large number of IoT devices and the implementation of the 6G system, it gives a possible risk that unauthorized users can pretend to be legitimate primary users and enter the spectrum in order to conduct malicious services. There is no alternative to resolving these security issues if network integrity and reliability are to be maintained.
2) Spectrum Sensing and Interference: The mobile nature of secondary users (SUs) within the sensing area significantly influences spectrum handoff; consequently, the process of spectrum sensing becomes inexact, and interference is likely to take place with primary users (PUs). Viable solutions are needed to relieve these two issues from the impacts they have on network performance.
3) Beamforming Efficiency: Beamforming at a single base station does not provide sufficient data that can be probed by the secondary users. Beamforming effectiveness brings network performance by achieving reliability in connectivity.

Towards this, the study is guided by the following objectives:

1) Develop an access policy to the spectrum that will ensure the highest levels of utilization possible without authorized or unauthorized interference from secondary users on the primary users and with minimal interference among themselves.
2) Develop a secure 6G base station based on ultra-massive MIMO to realize spectral efficiency, capacity improvement, and reliability in data transmission.
3) Allow effective beamforming techniques at the cognitive base station in the dynamic network environment and with varying user requirements, which will improve connectivity and network performance.
4) Design a secure routing protocol that guarantees safe data transmission, keeping the network from attacks and intrusions, thus ensuring the privacy and security of the 6G CIoT network.
5) Implement AI techniques in cognitive radio networks to enhance network adaptability, optimize resource allocation, and improve overall performance. AI will enable intelligent decision-making and more efficient communication among network devices, addressing challenges in the 6G CIoT environment.

Table 1 presents the list of abbreviations used throughout this paper.

TABLE 1. List of abbreviations.

Abbreviation	Full Form
6G	Sixth Generation
AI	Artificial Intelligence
IoT	Internet of Things
CRN	Cognitive Radio Networks
QKD	Quantum Key Distribution
PKI	Public Key Infrastructure
RL-ER	Reinforcement Learning-based Ensemble Regression
MLP-KF	Multi-Layer Perceptron with Kalman Filter
CSI	Channel State Information
MIMO	Multiple Input Multiple Output
ECC	Elliptic Curve Cryptography
QBO	Quantum-Inspired Bat Optimization
MBC	Master Blockchain Controller
DAHO	Dynamic Adaptive Hierarchical Optimization
SNR	Signal-to-Noise Ratio
RSSI	Received Signal Strength Indicator
PU	Primary User
SU	Secondary User
SHA-3	Secure Hash Algorithm 3
DSA	Digital Signature Algorithm
RL	Reinforcement Learning
ANN	Artificial Neural Networks
DLNN	Deep Learning Neural Network
LR	Logistic Regression
ACSS	Adaptive Compressed Spectrum Sensing
LOOCV	Leave-One-Out Cross Validation
MWC	Modulated Wideband Converter
N-OMP	Nearest Orthogonal Matching Pursuit
M2CSS	Multi-band Multi-user Cooperative Spectrum Sensing
MAP-MRF	Maximum a Posterior-Markov Random Field
6GCIoT	Sixth Generation Cognitive Internet of Things
XOR	Exclusive OR (Exclusive Disjunction)
CBS	Cognitive Base Station
DOA	Direction of Arrival
EQFA	Enhanced Quantum-Inspired Firefly Algorithm

II. LITERATURE SURVEY

The rapid evolution of wireless communication technologies has led to significant advancements in spectrum sensing techniques and strategies. This literature survey explores various methodologies and algorithms proposed in recent years, focusing on adaptive compressed spectrum sensing, cooperative strategies, machine learning applications, and security aspects in cognitive radio networks.

A. ADAPTIVE SPECTRUM SENSING TECHNIQUES

A low-cost adaptive compressed spectrum sensing (ACSS) method based on leave-one-out cross-validation (LOOCV) was proposed. The reconstructed signal was tested using one sample channel as a subset and was then tested again using various sampling channels. This method proved suitable for multiband spectrum sensing and effectively conserved sampling resources. However, the security level in ACSS was lower compared to other existing methods [19]. In the context of Imperfect Spectrum Sensing (ISS), primary channel statistics were introduced. ISS determines a set of closed-form expressions for the computed statistics as a function of the initial primary channel statistics. Novel estimators for the primary channel statistics were validated through simulations, providing accurate estimations based on ISS. Challenges arose in imperfect channel state information, the presence of multiple users on the same sub-channels, and interference among multiple primary cells [20]. Utilizing the modulated wideband converter (MWC) sub-Nyquist sampling structure, the nearest orthogonal matching pursuit (N-OMP) reconstruction technique was developed. It reduced computational complexity by up to 50%. However, the method's overall reconstruction accuracy suffered when applied to wideband spectrum sensing in low-signal-to-noise ratio (SNR) channels with significant sparsity [21], [22].

B. COOPERATIVE SPECTRUM SENSING STRATEGIES

A two-stage multi-band multi-user cooperative spectrum sensing (M2CSS) scheme was developed with the goal of allocating a portion of channels to secondary users (SUs) for spectrum sensing. An optimization problem was formulated to select a leader for each channel. This approach achieved high performance levels in both smaller and larger Cognitive Radio Networks (CRNs) compared to existing methods. SUs selected for sensing channels were constrained by energy limitations [23], [24]. Three cooperative sensing algorithms were proposed for centralized, clustered, and distributed secondary networks. Sensing collaboration was achieved by determining the highest posterior probability over the Markov random field. The Maximum a Posterior-Markov Random Field (MAP-MRF) framework and algorithms extended to CSS tasks with multiple PUs and SUs, offering higher computational efficiency and lower communication overhead than current approaches. One limitation was that the undirected signals of the framework were difficult to interpret using the model [25], [26].

C. MACHINE LEARNING APPLICATIONS IN SPECTRUM SENSING

A new adversarial learning-based spectrum sensing technique was developed to increase model adaptability. This method involved three interconnected neural networks (NNs), which deduced the spectrum status in a new test SNR set using characteristics derived from the less SNR-dependent universal features of the training SNR set. This approach achieved a lower spectrum sensing error rate compared to other classical signal processing-based and machine learning (ML) methods. A significant challenge was model generalization when applying ML to wireless communications [27]. Joint designs of digital filters and analog beam selection were developed at low cost. The adversarial learning-based spectrum sensing technique created three interconnected NNs, inferring the spectrum status in a new test SNR set using characteristics from the training SNR set's less SNR-dependent universal features. Compared to other methods, this technique exhibited a high error susceptibility [28], [29].

D. SECURITY ASPECTS IN COGNITIVE RADIO NETWORKS

A new adversarial learning-based spectrum sensing technique was introduced to enhance model adaptability. This methodology involved three interconnected NNs, which inferred the spectrum status in a new test SNR set using characteristics derived from the training SNR set's less SNR-dependent universal features. To mitigate the impact of false sensing reports by malicious users (MUs), a robust spectrum sensing technique was proposed. This technique utilized a double-sided neighbor distance (DSND) based on a genetic algorithm (GA) to filter out MU sensing reports in CSS. The GAMV-HDF model, a hard decision fusion model using GA majority voting, demonstrated more accurate and reliable simulation results. The presented scheme improved security accuracy performance but consumed more power compared to other methods [30].

A secure routing approach based on node behavior in CRNs was developed. Node behavior was quantified using a Belief Level (BL) parameter, and routing request and reply packets were encrypted using cryptographic methods. The algorithm's security functionality was confirmed and validated, but the higher probability of public parameter exploitation in cryptography introduced vulnerabilities [4]. For reliable cooperative ratio spectrum sensing in the CR-IoT, an ML-based malicious user detection model was constructed. The CR-IoT model employed a Support Vector Machine (SVM) algorithm to detect malicious users. The SVM's effectiveness in CR-IoT was verified with simulation results. However, the computational complexity was high, potentially causing packet drops during data transmission [31].

E. INTEGRATION OF COGNITIVE RADIO AND IOT

The basic functionalities of CR and Internet of Things (IoT) technology were integrated, presenting a five-layered framework for CR-enabled IoT. For the CR-based IoT architecture, the Robust Principal Component Analysis (RPCA-CN) designed a spectrum sensing algorithm. The model's detection efficiency was 10.7% higher and at least four times faster than earlier methods. However, the complexity resulting from the five-layered framework was a disadvantage [32].

III. PROBLEM STATEMENT

Cognitive Radio Networks (CRNs) have emerged as a solution to the challenge of spectrum scarcity in wireless communication. However, as we transition to 6G Cognitive IoT (6GCIoT), the demand for secure spectrum access and efficient beamforming becomes more critical, revealing shortcomings in current methodologies. The challenges in existing research include:

- **Mobility and Signal Strength:** Static mobility models in CRNs often result in reduced Received Signal Strength Indicator (RSSI), leading to significant data loss. If RSSI_{static} represents the signal strength with static mobility and $\text{RSSI}_{dynamic}$ with dynamic mobility, then the data loss due to static mobility is:

$$\Delta\text{RSSI} = \text{RSSI}_{dynamic} - \text{RSSI}_{static}$$

- **Spectrum Allocation:** Inefficient spectrum allocation strategies lead to underutilization of the spectrum, worsening scarcity issues. If S_{total} is the total available spectrum and S_{used} is the spectrum in use, the unused spectrum is:

$$S_{scarcity} = S_{total} - S_{used}$$

- **Scalability in CRNs:** Limited scalability in CRNs hinders the optimal use of idle spectrum bands. If $R_{current}$ denotes the current resources and R_{demand} represents the demanded resources, the resource inefficiency is:

$$\Delta R = R_{demand} - R_{current}$$

Given these challenges, this research aims to:

- Develop a dynamic mobility model to optimize RSSI values and minimize data loss.
- Create an intelligent spectrum allocation mechanism to reduce $S_{scarcity}$.
- Develop a framework to enhance CRN scalability and bridge the ΔR gap.

Incorporating these solutions will ensure improved Secure Spectrum Access and Beamforming in MIMO-6GCIoT, enhancing both security and performance.

IV. QUANTUM-SECURED IoT COMMUNICATION FRAMEWORK FOR COGNITIVE RADIO NETWORKS

This framework will be referred to as the Quantum-Secured IoT Communication Framework for Cognitive Radio Networks, and it starts with a dual-layer authentication procedure that integrates the Quantum Key Distribution mechanism simultaneously with Public Key Infrastructure. The approach draws on the inherently secure characteristics of quantum cryptography while utilizing established PKI protocols to ensure that IoT devices and secondary users are authenticated with maximum security and privacy guarantees. It authenticates first with biometric authentication hashed with the ID of IoT devices and users, followed by location information, using the hashing scheme SHA-3 (Keccak), which computes a unique and safe 256-bit hash value and stores it in the quantum-safe distributed ledger, which safely holds tamper-proof user information further from illegal access or unauthorized manipulation, thus enhancing security for CRNs based on IoT.

The dynamic Master Blockchain Controller is selected by the Dynamic Adaptive Hierarchical Optimization (DAHO) algorithm for efficient spectrum access and utilization. The DAHO chooses the MBC with the highest trust value, assuring that the node is most reliable to manage blockchain operations. Moreover, using the spectrum sensing ability to predict the availability of spectra, a novel Reinforcement Learning-based Ensemble Regression (RL-ER) model is presented.

In this regard, a Multi-Layer Perceptron-Kalman Filter (MLP-KF)-based system has been devised to predict Channel State Information (CSI) in a highly accurate and flexible manner for efficient communications in the IoT-based Cognitive Radio Networks (CRNs). This integrated approach has made the overall system highly secure, reliable, and operational in nature when deployed over a varying spectrum environment. The framework being proposed is shown in Figure 1.

A. AUTHENTICATION

At first, secondary users who need access to the spectrum are authenticated for a secure 6G communication link. In this proposed methodology, the secondary users are authenticated using a combination of quantum key distribution (QKD) and traditional public key infrastructure (PKI) to ensure maximum security and privacy.

B. VALID ID CREATION

For the authentication, the ID, biometric data, and location of each user are hashed to generate a valid ID. This process involves hashing the ID, biometric data, and location information using the SHA-3 (Keccak) hash function. SHA-3 operates on a 64-bit word, ultimately producing a 256-bit hash value. The compression of SHA-3 processes a state represented as a matrix and consists of three steps: initialization, absorption, and squeezing.

1) INITIALIZATION

During initialization, the state is filled with the 64-bit word as a chaining value $C_v = C_{v0}, \ldots, C_{v15}$, where $C_v \in S$ is a value of 1024 bits, salt values, and counter values $C_t = C_{t0}, C_{t1}$. The chaining value, salt value, and counter values are 64-bit words. Then, the 64-bit message is obtained from the secondary user information and is compressed as $val = val_0, val_1, \ldots, val_{15}$. Hence, the compression operation initially performed is given as:

$$\begin{bmatrix} val_0 & val_1 & val_2 & val_3 \\ val_4 & val_5 & val_6 & val_7 \\ val_8 & val_9 & val_{10} & val_{11} \\ val_{12} & val_{13} & val_{14} & val_{15} \end{bmatrix} \tag{1}$$

$$\downarrow$$

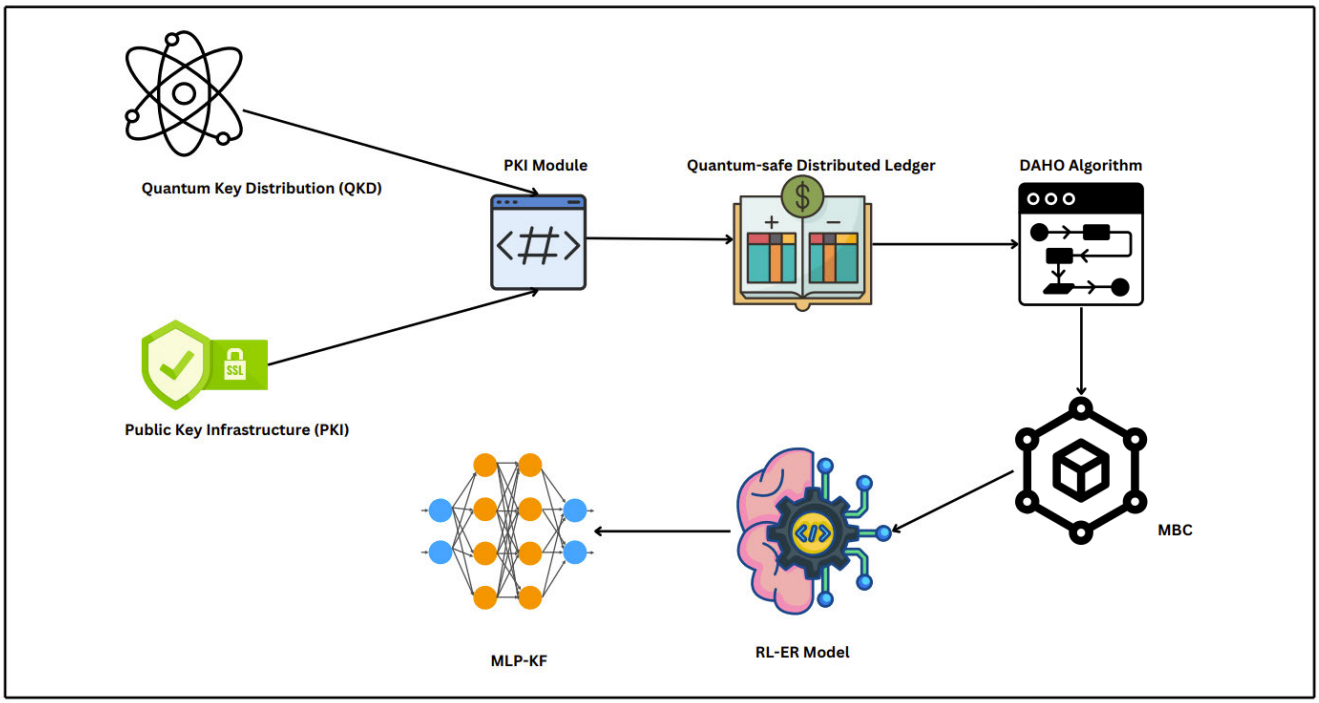

FIGURE 1. Quantum-Secured IoT communication framework for Cognitive Radio Networks (CRNs).

$$\begin{bmatrix} CV_0 & CV_1 & CV_2 & CV_3 \\ CV_4 & CV_5 & CV_6 & CV_7 \\ Sa_0 \oplus Con_0 & Sa_1 \oplus Con_1 & Sa_2 \oplus Con_2 & Sa_3 \oplus Con_3 \\ Ct_0 \oplus Con_4 & Ct_1 \oplus Con_5 & Ct_0 \oplus Con_6 & Ct_1 \oplus Con_7 \end{bmatrix} \tag{2}$$

2) ABSORPTION

The compression function then absorbs the input data into the state through XOR operations, followed by non-linear transformations.

3) SQUEEZING

After the absorption phase, the state undergoes a squeezing phase to produce the final hash value. The new chaining values $\bar{C_v} = \bar{C_{v1}}, \bar{C_{v2}}, \ldots, \bar{C_{v15}}$ are obtained, producing the hashed value for each user by using the initial chaining value and the salt value, which is represented as:

$$\overline{C_v} = C_v \oplus S_a \oplus \sum_{ini=0}^{7} val_{ini} \oplus \sum_{fin=8}^{15} val_{fin}$$

Here, $\overline{C_v}$ depicts the new chaining (hashed) value, $\oplus$ depicts the compression operation, and ini, fin depict the initial and final set of compressed values. These hashed details of the user are stored in a quantum-safe distributed ledger, which is mathematically expressed as:

$$B = [b_1, b_2, \ldots, b_a] \quad \text{or} \quad b_z,\ z = 1, 2, \ldots, a$$

where B denotes the distributed ledger with b_a blocks that contain hashed information of the primary and secondary users.

C. SELECTION OF THE MBC

In the existing blockchain technology, decentralized control is typically given to a user randomly. This random assignment can make the network vulnerable to threats if a single peer block is attacked. To mitigate this vulnerability, we propose selecting the Master Blockchain Controller (MBC) using the Dynamic Adaptive Hierarchical Optimization (DAHO) algorithm.

The initial population is considered as the created peer blocks. The fitness of each individual is calculated to assign roles. The individual with the best fitness is selected as the leader. Here, fitness is considered as the user with the highest trust value. Thus, the user with the highest trust value will get the MBC.

1) POSITION UPDATE

After calculating the fitness of each individual, the position is updated according to the obtained fitness. The position update of individuals combines hierarchical optimization with adaptive strategies. The position update is formulated as follows:

$$v_i(t+1) = w \cdot v_i(t) + \gamma \cdot (p_{hi} - x_i(t)) + \delta \cdot (p_{gi} - x_i(t)) \tag{3}$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{4}$$

where $v_i(t)$ denotes the velocity of individual i at time t, $x_i(t)$ denotes the position of individual i at time t, w is the inertia weight, γ and δ are hierarchical and global learning factors respectively, p_{hi} is the hierarchical best position of individual i, and p_{gi} is the global best position found by the population.

2) INITIALIZATION

The parameters required for the DAHO algorithm are initialized as follows:

- Inertia weight w.
- Hierarchical learning factor γ.
- Global learning factor δ.
- Population size.
- Maximum number of iterations.

3) FITNESS EVALUATION

The fitness of each individual is evaluated based on its trust value. The individual with the highest trust value is identified as the best candidate for MBC.

4) VELOCITY AND POSITION UPDATE

The velocity and position of each individual are updated according to the equations given above. Hierarchical learning allows individuals to learn from the best solutions in their hierarchical structure, while global learning allows them to learn from the overall best solution. The process is iterated until the maximum number of iterations is reached or convergence criteria are met.

This work introduces a new method to select the main controller in a blockchain by using the Dynamic Adaptive Hierarchical Optimization (DAHO) algorithm. Unlike random selection, this method evaluates the trustworthiness of nodes and assigns control to the most reliable node. This enhances the security and robustness of the system. The DAHO algorithm's combination of hierarchical and adaptive learning dynamics makes it highly effective for dynamic and distributed environments like blockchain networks.

Algorithm 1 Dynamic Adaptive Hierarchical Optimization (DAHO) for MBC Selection

Require: Peer blocks
Ensure: Master Blockchain Controller (MBC)

```
Begin
Initialize individuals x_i and velocities v_i
Evaluate fitness of each individual
Set hierarchical best p_hi and global best p_gi
Set iteration counter t = 0
while t ≤ max iterations do
    for each individual i do
        Update velocity v_i(t+1) using v_i(t+1) = w·v_i(t) + γ·(p_hi − x_i(t)) + δ·(p_gi − x_i(t))
        Update position x_i(t+1) using x_i(t+1) = x_i(t) + v_i(t+1)
        Evaluate fitness of x_i(t+1)
        if fitness of x_i(t+1) is better than p_hi then
            Update p_hi = x_i(t+1)
        end if
        if fitness of x_i(t+1) is better than p_gi then
            Update p_gi = x_i(t+1)
        end if
    end for
    Increment iteration counter t = t + 1
end while
return MBC p_gi
End
```

D. SPECTRUM ACCESS

The MBC should get sufficient spectrum access. Hence, the proposed method utilized two methods for spectrum access: spectrum sensing and Channel State Information (CSI) prediction.

1) SPECTRUM SENSING

Spectrum sensing is a vital process in the CRN because the CBS provides spectrum access to the unlicensed secondary user when the spectrum is not accessed by the primary user. Hence, for the secondary user α, the CBS senses which spectrum is available and ready to be used by the user α. To sense the spectrum availability, a novel Reinforcement Learning-based Ensemble Regression (RL-ER) model is proposed. The sensing of the spectrum depends on the Signal Noise Ratio (SNR), noise level, and trust.

a: INPUT

The several spectrums used by the primary users are taken as the input for the RL-ER model and the input (spectrum bands) are represented as $(In_1, In_2, .., In_z)$ or In_{spe}, In represents the spectrum band in the CBS and the weight values are represented as $(N_1, N_2, .., N)$ or N_{spe}. These spectrum inputs are prepared for the further process in the input layer of the ensemble regression.

b: HIDDEN LAYERS

In the RL-ER model, multiple hidden nodes are employed. The hidden layers process the input layer neurons and the weight values to produce the required spectrum based on the sensing requirements, such as Signal to Noise Ratio (SNR) and trust. The hidden node calculation is given as follows:

$$Hid_{\Gamma} = \sum_{spe1}^{\Gamma} (In_{spe} \times N_{spe}) \tag{5}$$

Here, the hidden nodes corresponding to the input and weight values are Hid_{Γ}. Then, the ensemble regression activation function is applied.

c: ACTIVATION FUNCTION

The activation function of ensemble regression is to convert the hidden node outcome into a prediction of spectrum

availability. The ensemble method combines the predictions from multiple regression models to improve accuracy. The steps of the ensemble regression activation function are given as follows:

The ensemble regression is mathematically expressed as:

$$pr(\Omega_\Gamma = 1|Hid_\Gamma) = \nu[Hid'_\Gamma \lambda] = \frac{e^{(Hid'_\Gamma \lambda)}}{1 + e^{(Hid'_\Gamma \lambda)}} \quad (6)$$

where, $pr(\Omega = 1)$ denotes the probability of Hid_Γ node predicting the spectrum as available (class 1), (Ω) denotes the coefficient associated with the hidden node with a value of 0 or 1, ν depicts the activation function. $\lambda = \lambda_1, \lambda_2, \ldots .\lambda_u$ is the estimated coefficient. Then, the log-likelihood function is expressed as:

$$L_i(\lambda) = -\sum_{\Gamma=1}^{\tau} \Big[\Omega_\Gamma \cdot \log\big(\nu(Hid'_\Gamma \cdot \lambda)\big) + (1 - \Omega_\Gamma) \cdot \log\big(1 - \nu(Hid'_\Gamma \cdot \lambda)\big)\Big] \quad (7)$$

The coefficient λ is obtained by minimizing the log-likelihood function $L_i(\lambda)$. However, in high-dimensional applications like spectrum sensing, the model may lead to an overfitting problem. Hence, a sparse regularization term is added to the equation, which is given as:

$$\lambda = \arg\min\left(L_i(\lambda) + K \cdot \sum_{\Gamma=1}^{\tau} \mathrm{reg}(\lambda_\Gamma)\right) \quad (8)$$

where an adjusting parameter with only positive values is K, $\mathrm{reg}(\lambda_\Gamma)$ depicts the regularization item. Finally, the cost function using Reinforcement Learning is proposed to enhance the sensing gain of the RL-ER model. The RL uses action values to improve the behavior of the ensemble regression model, which is given as:

$$M(\hat{S}t\lambda, AC\lambda) = M(St_\lambda, Ac_\lambda) + Lr\Big[d(St, Ac) + \gamma M(St'_\lambda, Ac'\lambda) - M(St\lambda, Ac_\lambda)\Big] \quad (9)$$

where, $\hat{M}(\cdot)$ depicts the new Q-value based on the current state of the spectrum St_λ, and current action Ac_λ, of the accessible spectrum with respect to the spectrum sensing policy. St' and A'_c represents the maximum expected next state and the next action of the agent using the spectrum respectively. Lr denotes the learning rate to update the estimation of $M(St, Ac)$, $d(\cdot)$ depicts the reward from the environment for taking action in the state St and γ represents the discount factor for future rewards.

Hence, the RL-ER model predicts the spectrum of the primary user ready to be used by the secondary user as class 1. And if no spectrum is available for use, then the classifier returns class 0.

E. CSI PREDICTION

The available output spectrum predicted by the RL-ER model is ready to be accessed by the secondary user, hence the Channel State Information (CSI) of the spectrum channel M is predicted for efficient 6G communication. The CSI detects how the signal propagates between the transmitter and the receiver with the presence of interference, fading, and power decay with respect to time. The MLP-KF algorithm plays a crucial role in predicting CSI, which is key to achieving efficient transmission. This algorithm is vital for effective transmission as it leverages the learning strengths of a Multi-Layer Perceptron (MLP) alongside the precise estimation capabilities of a Kalman Filter. The MLP processes the input data, generating initial predictions, which are subsequently refined by the Kalman Filter to improve the accuracy of the CSI estimation. The process is carried out as follows:

1) MLP INITIALIZATION

The MLP is first initialized with input data from the spectrum channel, which it processes through its hidden layers to produce initial CSI predictions.

2) KALMAN FILTER REFINEMENT

The next step is fine-tuning these initial predictions for the effect of noise and channel variations. The updating process of the state equation in the Kalman Filter uses the previously estimated CSI H_t, the predicted CSI at time $t + 1$, and the measured CSI M_{t+1}. The Kalman gain K_t is updated as follows:

$$K_t = \frac{\Phi_t}{\Phi_t + R} \quad (10)$$

where Φ_t denotes the error covariance and R represents the measurement noise covariance. The error covariance Φ_t is updated as:

$$\Phi_{t+1} = (1 - K_t)\Phi_t + Q \quad (11)$$

where Q is the process noise covariance.

The integration of MLP with the Kalman Filter further increases the accuracy and adaptability of CSI predictions, thereby enhancing the efficiency of spectrum utilization in 6G communication networks.

This method introduces a novel approach for spectrum sensing using the RL-ER model and CSI prediction using the MLP-KF algorithm. The combination of reinforcement learning and ensemble regression improves the accuracy and robustness of spectrum sensing, while the integration of MLP and Kalman Filter provides precise and adaptive CSI predictions, ensuring efficient spectrum access and utilization in 6G communication networks.

F. SECURE BEAMFORMING

Based on the CSI, the quality of the signal needed to be transmitted by the antenna is controlled using cooperative beamforming. Here, along with the CSI value, beam score, array factor, and Direction Of Arrival (DOA) are also considered for secure beamforming. Assuming that each relay node received the data from the source node, the μ relay

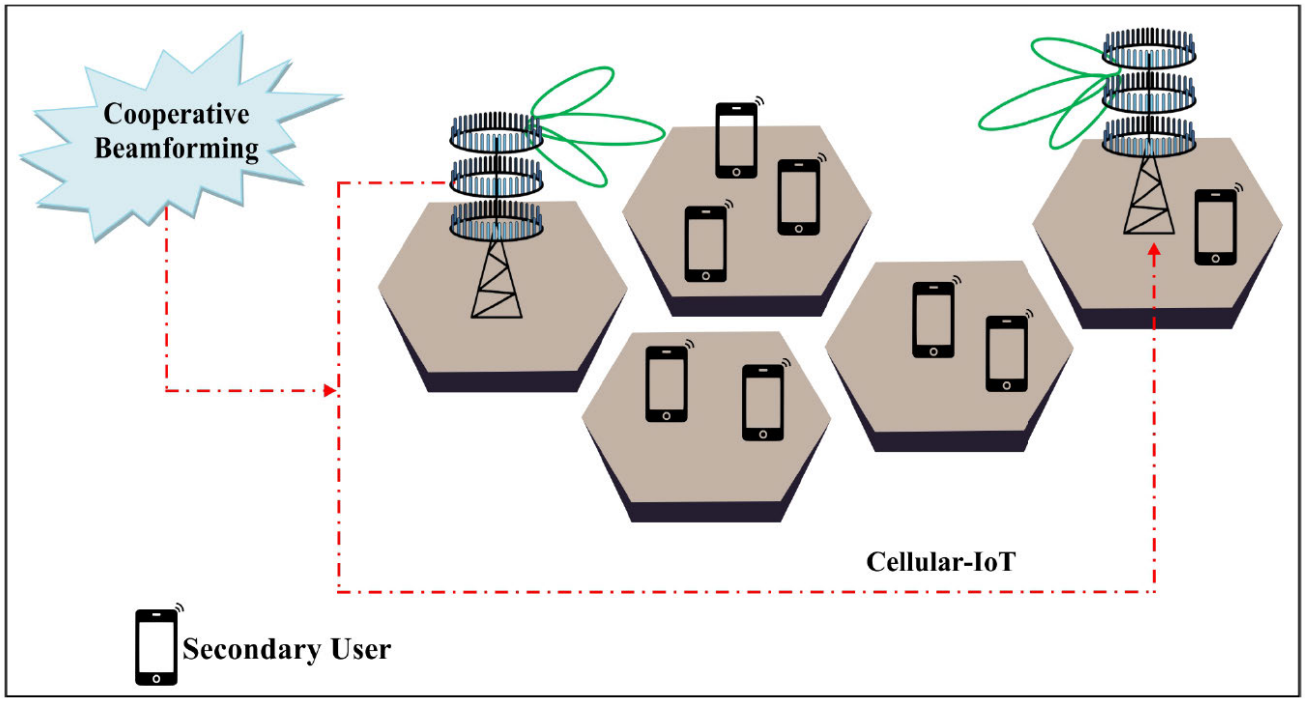

FIGURE 2. Beamforming model.

nodes form a virtual link to the destination. The cooperative beamforming model is given in Figure 2.

The energy consumed during the secure beaming and delivering of the packet to the source node is given by,

$$G_s = \left(\frac{P_{\text{ow_req}}}{\ell}\right)(1+\mu)P_{\text{ow}}\ell T_r \tag{12}$$

where, G_s denotes the consumed energy during the beamforming.

$P_{\text{ow_req}} = \frac{\omega_{th} \times \text{bit_px} \times P_{\text{ow}}}{\|\Phi\|^2}$ is the required power to reach the target with threshold ω_{th}.

$T_r = \frac{\ell}{\text{bit_px} \times X}$ is the total transmission time to send the l bits, and bit_px depicts the number of bits per symbol for the secure beamforming in the CRN.

G. ROUTE SELECTION

After secure beamforming, the shortest route for the transmission of the data to the user is determined using the Enhanced Quantum-Inspired Firefly Algorithm (EQFA). This algorithm is efficient in finding the shortest path between the source node and destination node for sending $'l'$ bits.

1) INITIALIZATION

The EQFA starts by initializing the positions of fireflies (nodes) randomly within the network. The initial brightness of each firefly is calculated based on the distance to the destination node.

2) MOVEMENT AND ATTRACTION

The movement of fireflies is influenced by their attractiveness, which is determined by their brightness. Fireflies with higher brightness attract others, causing them to move towards the brighter fireflies. The attractiveness β is calculated as:

$$\beta = \beta_0 \cdot e^{-\gamma r^2} \tag{13}$$

where β_0 is the initial attractiveness, γ is the light absorption coefficient, and r is the distance between two fireflies.

3) QUANTUM-INSPIRED UPDATE

To enhance exploration, quantum-inspired principles are applied. Each firefly updates its position using a probability amplitude function, allowing it to explore new positions in the solution space:

$$x_i(t+1) = x_i(t) + \beta(x_j(t) - x_i(t)) + \alpha \cdot (rand - 0.5) \tag{14}$$

where $x_i(t)$ and $x_j(t)$ are the positions of firefly i and j at time t, α is the step size, and $rand$ is a random number.

4) CONVERGENCE

The process continues until the fireflies converge to the brightest firefly, which represents the optimal or near-optimal path. The selected shortest path is represented as *Path*.

H. SECURE ROUTING (COMMUNICATION)

After finding the shortest path, the base station transfers the data via *Path* securely using the proposed Quantum-Inspired

Bat Optimization with Elliptic Curve Cryptography (QBO-ECC) algorithm. In the conventional ECC algorithm, the keys are generated randomly, which could cause severe security problems. Hence, to solve this problem, the keys are selected using the Quantum-Inspired Bat Optimization (QBO) algorithm.

The Elliptic Curve is defined over a finite field F, and its equation is:

$$y^2 = x^3 + ax + b \tag{15}$$

where a and b are constants that define the curve.

The steps for secure communication using the QBO-ECC algorithm are as follows:

1) KEY GENERATION

The initial population of bats is considered the set of potential keys. Each bat's position represents a possible key. The fitness of each key is evaluated based on its ability to encrypt and decrypt the data securely.

2) MOTION OF BATS

Bats move towards the best solution using echolocation and random walks. The velocity and position update equations are:

$$v_i(t+1) = v_i(t) + (x_i(t) - x_*) \cdot f \tag{16}$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{17}$$

where $v_i(t)$ and $x_i(t)$ are the velocity and position of bat i at time t, x_* is the current global best position, and f is a random frequency value.

3) FREQUENCY AND LOUDNESS UPDATE

The frequency and loudness of each bat are updated dynamically to balance exploration and exploitation:

$$f_i = f_{\min} + (f_{\max} - f_{\min}) \cdot rand \tag{18}$$

$$A_i(t+1) = \alpha \cdot A_i(t) \tag{19}$$

where $f_{\min}$ and $f_{\max}$ are the minimum and maximum frequency values, $A_i(t)$ is the loudness of bat i at time t, and α is a constant.

4) SELECTION OF BEST KEY

The bat having maximum fitness is used as the best key for encryption. This key would be used to encrypt the data before transmission.

5) ENCRYPTION AND TRANSMISSION

Using the selected key, the data is encrypted and sent securely. The encryption process is given by:

$$C = P \cdot K \tag{20}$$

where C is the ciphertext, P is the plaintext, and K is the key.

6) DECRYPTION

The receiver uses the corresponding private key to decrypt the data:

$$P = C/K \tag{21}$$

Hence, the data is securely transmitted in the 6G CRN using the QBO-ECC algorithm. The working procedures of the proposed secure communication algorithm QBO-ECC are given as follows:

Algorithm 2 QBO-ECC

Require: t bits
Ensure: Encrypted data

```
Begin
Initialize bats' positions and velocities
Evaluate fitness of each bat
Set iteration Iter = 1
while Iter ≤ Iter_max do
    for each bat do
        Update frequency, velocity, and position
        Evaluate fitness of new position
        if new position is better than current best then
            Update current best
        end if
    end for
    Select the bat with the highest fitness as the best key
    Encrypt data using the best key
    Update iteration counter Iter = Iter + 1
end while
Return encrypted data
```

The technique proposed is characterized by its new way of implementing Quantum-Inspired Bat Optimization (QBO) algorithm beside of Elliptic Curve Cryptography to enhance the security of data transmission. The conventional ECC algorithms are generally produced by means of a random key generator that creates a potential vulnerability in the security. To this direction, the QBO algorithm is employed site-wise which is guided by the bat's echolocation. In this bio-mimetic approach, the QBO strategy ensures survival by modeling both an exploitation (attack) phase and an exploration (cruise) phase that reminds the hunting tactics of the bat. The mechanism of these two phases strengthens the security of the key generation. Furthermore, its successful deployment in 6G CRN indicates its qualification to deal with the increasingly sophisticated security requirements in the onboard of emerging communication networks.

V. RESULTS AND DISCUSSION

In this section, an analysis is conducted on the empirical findings of the proposed framework by comparing it with established methodologies. The experiments were performed using the *OMNeT++* software in conjunction with the *INET* framework, establishing a comprehensive and flexible simulation environment which enabled a meticulous assessment of the proposed design. The subsequent sections will

expound on the comparisons of results concerning various performance metrics.

A. PERFORMANCE ANALYSIS OF PROPOSED RL-ER

Within this segment, the effectiveness of the proposed Reinforcement Learning-based Ensemble Regression ($RL - ER$) technique is substantiated by contrasting outcomes with prevailing Deep Learning Neural Networks ($DLNN$), Logistic Regression (LR), and Artificial Neural Networks (ANN) models. The validation of results is conducted based on parameters such as prediction accuracy, computational efficiency, and error rates.

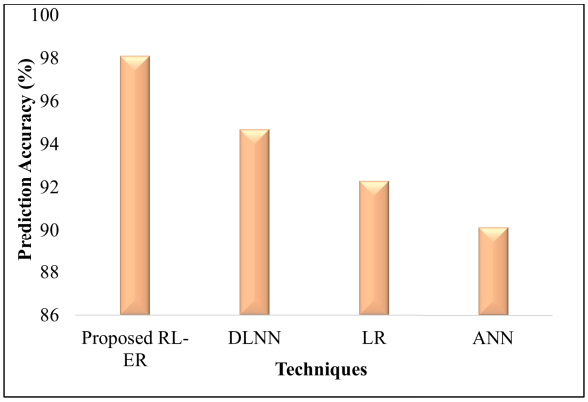

FIGURE 3. Prediction accuracy of RL-ER with DLNN, LR, and ANN.

Figure 3 illustrates the performance measure of the proposed $RL - ER$ in terms of prediction accuracy. The proposed technique achieves a higher prediction accuracy of 98.1%. In contrast, conventional methods such as $DLNN$, LR, and ANN show lower prediction accuracies of 94.7%, 92.3%, and 90.1%, respectively. Hence, it is evident that the proposed methodology excels in prediction accuracy.

TABLE 2. Comparative analysis of proposed RL-ER based on time consumption.

Techniques	Time Consumption (ms)
Proposed RL-ER	5127
DLNN	8235
LR	10149
ANN	11973

Table 2 shows the time consumption of the proposed and existing methods. Time consumption measures the time required to process the data. From Table 2, the proposed technique consumes 5127ms, which is significantly lower compared to the existing DLNN (8235ms), LR (10149ms), and ANN (11973ms) techniques. Thus, it is clear that the proposed system outperforms other conventional methodologies in terms of computational efficiency.

B. ERROR RATE ANALYSIS OF RL-ER

Error rate is a critical metric for evaluating the performance of predictive models. It is defined as the ratio of the number of incorrect predictions to the total number of predictions.

$$\text{Error Rate} = \frac{\text{Number of Incorrect Predictions}}{\text{Total Number of Predictions}}$$

To assess the error rate of $RL - ER$, we tested the model on a set of data samples with known outcomes and compared the $RL - ER$ predictions with the actual outcomes.

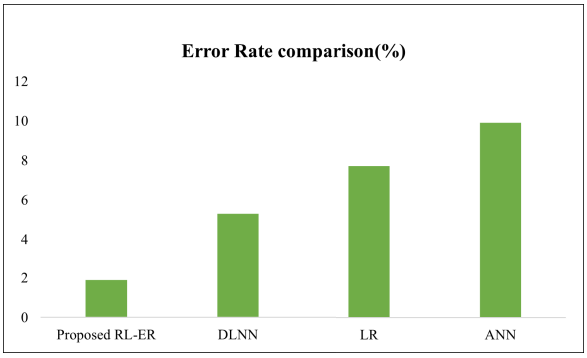

FIGURE 4. Error rate comparison of RL-ER with DLNN, LR, and ANN.

Figure 4 illustrates a comparison of error rates among various methodologies. The $RL - ER$ method proposed managed to attain the minimum error rate of 1.9%, surpassing $DLNN$, LR, and ANN, which exhibited error rates of 5.3%, 7.7%, and 9.9%, respectively. These findings suggest that $RL - ER$ notably diminishes the error rate in comparison to other techniques. This enhancement can be ascribed to the novel incorporation of Reinforcement Learning with ensemble regression within the $RL - ER$ framework, augmenting the model's capacity to adjust and glean insights from diverse data patterns, consequently leading to more precise forecasts.

TABLE 3. Comparative analysis of proposed RL-ER based on prediction accuracy.

Techniques	Prediction Accuracy (%)
Proposed RL-ER	98.1
DLNN	94.7
LR	92.3
ANN	90.1

Table 3 provides a comparison of the prediction accuracy between the proposed RL-ER method and existing models. The data clearly indicates that RL-ER consistently outperforms DLNN, LR, and ANN, not only in prediction accuracy but also in computational efficiency and error rates. These advantages make RL-ER an exceptional choice for predictive modeling tasks.

C. PERFORMANCE MEASUREMENT OF PROPOSED QBO-ECC

The performance of the proposed Quantum-Inspired Bat Optimization with Elliptic Curve Cryptography ($QBO - ECC$) technique is verified by analyzing the result parameters, such as encryption time, decryption time, and security

level. Further, these outcomes are compared with some of the existing techniques, such as Elliptic Curve Cryptography (*ECC*), Digital Signature Algorithm (*DSA*), and Elgamal.

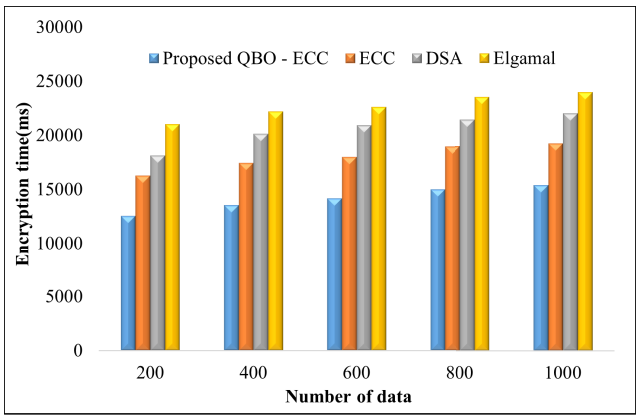

FIGURE 5. Performance measurement based on encryption time.

Figure 5 illustrates the time taken by the proposed $QBO - ECC$ method to encrypt the data. Lesser encryption time indicates better performance. As the size of the data increases, the encryption time also increases. For a minimum number of data (200), the encryption time of the proposed method is 12567ms. For a maximum number of data (1000), the encryption time is 15342ms. On the other hand, for 200 data points, the existing *DSA* has an encryption time of 18124ms, which is higher compared to the proposed $QBO - ECC$ technique. Likewise, the encryption time varies for other existing techniques. Therefore, it is clear that the proposed model encrypts the data in less time than other methods.

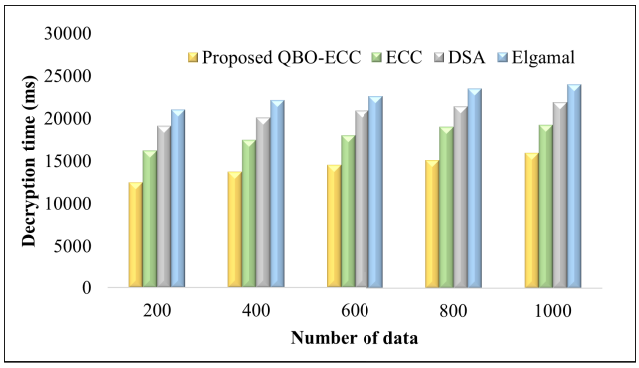

FIGURE 6. Performance measurement based on decryption time.

The decryption time comparison is shown in Figure 6. The decryption time measures the time taken to decrypt the data after encryption. Similar to encryption time, less decryption time indicates better performance. The decryption time of the proposed method by varying the number of data points is 12453ms (200), 13691ms (400), 14498ms (600), 15034ms (800), and 15874ms (1000). On the contrary, the existing *ECC* has decryption times of 20761ms (200), 21532ms (400), 23873ms (600), 25302ms (800), and 30442ms (1000). In the same way, the decryption time of other existing *DSA* and Elgamal techniques varies. Thus, it is evident that the proposed $QBO - ECC$ method is more efficient than other state-of-the-art models.

TABLE 4. Key generation time analysis for various techniques.

Techniques	Key Generation Time (ms)
Proposed QBO-ECC	235
ECC	400
DSA	380
Elgamal	420

D. KEY GENERATION TIME ANALYSIS

From Table 4, the proposed $QBO - ECC$ algorithm demonstrates a more efficient key generation process, taking only 235ms to generate the required keys. In contrast, established methods such as ECC, DSA, and Elgamal take considerably longer, with *ECC* taking 400ms, *DSA* taking 380ms, and Elgamal requiring 420ms. The efficiency in key generation by the $QBO - ECC$ can be attributed to its innovative use of the Quantum-Inspired Bat Optimization algorithm, which streamlines the process of key selection. A faster key generation time not only indicates a more efficient encryption system but also suggests that the system can quickly adapt and regenerate keys, enhancing the overall security framework. In dynamic environments, where keys might need to be generated or changed frequently, such efficiency can be invaluable. The reduced key generation time of the proposed $QBO - ECC$, when compared to other techniques, solidifies its position as a superior and efficient cryptographic method for secure data transmission. The key generation time is a crucial performance metric for encryption systems. It represents the amount of time taken by an encryption algorithm to generate the necessary keys for encryption and decryption processes.

VI. CONCLUSION

The proposed framework enhances security and efficiency in next-generation wireless systems to a significant extent. Through the amalgamation of Quantum Key Distribution with Public Key Infrastructure, it establishes a robust dual-layer authentication system. The model founded on Reinforcement Learning-based Ensemble Regression (RL-ER) achieves an accuracy of 98.1% and an error percentage of 1.9%, exceeding current methodologies. Furthermore, the Quantum-Inspired Bat Optimization with Elliptic Curve Cryptography (QBO-ECC) diminishes encryption time to 12,567 ms and decryption time to 12,453 ms, along with a rapid key generation time of 235 ms. These outcomes showcase the framework's superior performance in secure spectrum access, efficient beamforming, and scalable communication. Future research will delve into additional AI improvements and practical implementations to ensure resilience in diverse settings.

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P. DEEPANRAMKUMAR received the Ph.D. degree in computer science from Vellore Institute of Technology (VIT), Vellore, India. He is an Assistant Professor (Senior) with the School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India. His research interests include cognitive radio networks, the Internet of Things (IoT), the Internet of Vehicles (IoV), wireless sensor networks, and deep learning.

A. HELEN SHARMILA received the Ph.D. degree in computer science from Vellore Institute of Technology (VIT), Vellore, India. She is an Assistant Professor (Senior) with the School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India. Her research interests include the Internet of Things (IoT), wireless body area networks, wireless sensor networks, and deep learning.